

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

ANSWERS & COMMENTS (18 Sept)

BIOLOGY
CLASS

2bi2_____

REGISTRATION
NUMBER

Biology

Paper 4 Practical

9744/04

18 August 2022

2 hours 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, Biology class, and registration number on all the work you hand in.
Give details of the practical shift and laboratory in the boxes provided.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your workings or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part of question.

Shift			
1	2	3	
Laboratory			
BI23	BI24	CM43	CM42

For Examiner's Use	
1	25
2	30
Total	55

This document consists of **16** printed pages and **4** blank pages.

Answer **all** questions.

- 1 You are required to investigate the effect of different concentrations of bromelain on gelatin breakdown by performing a serial dilution.

Bromelain is a proteolytic enzyme, which catalyses the hydrolysis of proteins into their amino acid building blocks. This enzyme can be found in fruits such as pineapple.

Gelatin contains structural proteins derived from animal tissues high in collagen. It sets into a gel-like structure upon cooling after dissolving in warm water. The action of bromelain on gelatin gel results in liquification of the gel.

Proceed as follows.

- 1 Weigh 4 g of gelatin powder **G** in a 250 ml beaker.
- 2 Dissolve the gelatin powder **G** in 100 cm³ of hot water (approximately 60 °C).
- 3 Pour the mixture into 5 separate large specimen tubes **L**.
- 4 Place the specimen tubes in an **ice water bath** for at least 15 minutes to allow setting of gelatin.
- 5 Check the consistency of the gelatin by tilting the mixture slightly. If the gelatin does not move, the gelatin gel is ready.

Complete the preparation for setting of gelatin gel before proceeding further.

- (a) (i) Using the information provided, describe a method to study effect of different concentrations of bromelain on gelatin breakdown.

In your plan, you must use:

- gelatin gel prepared in step 5
- pineapple sample, labelled **P**
- mortar and pestle
- small specimen tubes, labelled **S**
- tea filter bag
- syringes

Your plan method should:

- have a clear and helpful structure so that the method described could be repeated by anyone reading it
- include details to ensure that results are accurate and repeatable as possible
- identify the dependent and independent variables
- use the correct technical and scientific terms
- only make use of the apparatus and materials provided

Marking Point		Guidance
A.	Independent variable	Independent variable: different concentration of bromelain / pineapple juice + 5 values; Dependent variable: height of gelatin / volume of liquid collected / time taken for gelatin gel to liquify (completely or to a named extent)
B.	accurate and repeatable	Perform 2 more replicates AND calculate mean or identify/eliminate/remove anomalies; Note: not awarded if volume stated is insufficient for repeats as stated.
C.	Material and apparatus	Students used all the “must use” apparatus and not other apparatus not provided Names of apparatus should follow what is stated in the question. Note: award mark as long as reasonable clarity in naming of material and apparatus
D.	Dilution table	Show dilution table, using serial dilution. R: simple dilution Note: only marking for correct volumes. no penalty for less than ideal H/P
E.	Method to correctly collect dependent variable	Dependent variable: height of gelatin / volume of liquid collected / time taken for gelatin gel to liquify (completely or to a named extent)
F.	Method to prepare juice extract	
G.	Reference to a suitable controlled variable	Mass of pineapple used (weighing of pineapple) Volume / height of gelatin gel in the specimen tube Temperature using waterbath Volume of pineapple juice added
H.	Define concentration of bromelain / pineapple juice	E.g. 100% concentration or 1 arbitrary unit pineapple juice

Suggested Procedure

1. Weigh 20g of pineapple using a weighing balance
2. Crush the slice of pineapple into pulp using pestle and mortar
3. Add in 5 cm³ of water and crush for another 1 minute
4. Pour the mixture through the tea filter bag and extract the pineapple juice
5. Collect the filtrate in a beaker.
6. Perform serial dilution based on the table below.

Concentration of bromelain solution OR pineapple juice / %	Volume of solution from previous concentration / cm ³	Volume of DI water / cm ³	Total volume / cm ³
100.0	5.0	0.0	5.0
50.0	5.0	5.0	5.0
25.0	5.0	5.0	5.0
12.5	5.0	5.0	5.0
6.25	5.0	5.0	5.0

7. Once the gelatin set, label the different specimen tube with the different concentration of bromelain solution used using a marker.
8. Using the specimen tube for 100% bromelain solution, add in 5 cm³ of bromelain solution and start the stopwatch immediately.
9. At the end of 10 minutes, pour the liquified gelatin with the bromelain solution into a specimen tube and measure the volume collected using the 5ml and 1ml syringes.
10. Record the volume collected in the table below.
11. Repeat step 10 to 11 for the rest of the bromelain solution.

- 6 Carry out your method described in (a)(i) to investigate the effect of different concentration of bromelain on gelatin breakdown. You do **not** need to carry out repeats.

Record your results in a suitable format in (a)(ii).

- (ii) Use this space to record, in a suitable format, your results from step 6.

H – correct header (minimally: concentration of bromelain solution/ % + [3]
volume collected/ cm³ or equivalent)

Note: if student writes volume of gel collected, the volume collected must be <1.0cm³

Accept if unit for concentration of the bromelain unit is defined as x or arbitrary units.

P – precision [dp for volume: 1 dp, 2 dp (smallest unit of 1.0ml 0.01ml)]
Consistency

T – mark for probable trend (accept if trend not strictly decreasing)

C – not assessed

[Reject: text/ numbers written in pencil considered to be 'invisible']

Concentration of bromelain solution / %	Volume of P (5cm ³) + G collected in 10 minutes/ cm ³		
100	5.60	5.65	5.55
50	5.25	4.70	5.50
25	5.10	5.38	5.45
12.5	5.00	5.63	5.30
6.25	5.00	5.85	5.00

[Accept T if volume (G) collected is <1.0cm³]

Concentration of bromelain solution / %	Volume of P (2cm ³) + G collected in 15 minutes/ cm ³		
100	2.45		
50	2.40		
25	2.20		
12.5	2.10		
6.25	2.10		

[Accept T if volume (G) collected is <1.0cm³]

Concentration of bromelain solution / %	Initial mass of gel + specimen tube /g	Final mass of gel + specimen tube after 10 minutes/ g	Change in mass of gel after 10 minutes/ g
100	31.1	30.6	- 0.5
50	30.6	30.4	- 0.4
25	28.9	28.8	- 0.1
12.5	32.0	31.9	- 0.1
6.25	28.6	28.5	- 0.1

Concentration of bromelain solution / %	Time taken to liquefy completely/ minute	Time taken to liquefy completely/ s		
100	27	1620		
50	31	1860		
25	30	1800		
12.5	33	1980		
6.25	34	2040		

Results achieved by tilting the gel at minute interval, until all liquefied gel can flow without 'lumps'. Limitations of method: gel integrity is compromised with each tilt to check.

[Accept T if time for gel to liquefy completely range from 1500 – 2100s]

Concentration of bromelain solution / %	Initial height of gel in specimen tube /cm	Final height of gel in specimen tube /cm	Change in height of gel in specimen tube /cm
100	3.0		
50	2.9		
25	3.0		
12.5	2.9		
6.25	2.9		

Considerations:

External diameter of specimen tube ~3.1cm. Internal diameter of specimen tube ~3.0cm. Approximate cross-sectional area of specimen tube = ~5.39cm². Given a 1cm³ change in volume of gel, the height change is less than 0.2cm/ 2mm.

Boundary between gel and pineapple extract is difficult to determine.

[Accept T if change in height < 2mm]

Reject fractions or unknowns in data for concentration:

Concentration of bromelain solution / %	Volume collected / cm ³
$\frac{1}{2}x$	6.2
$\frac{1}{4}x$	5.8
$\frac{1}{8}x$	5.2
$\frac{1}{16}x$	5.0
$\frac{1}{32}x$	5.0

(iii) State and explain one variable that is kept constant in this investigation.

1. State: Concentration of gelatin mixture used / mass of gelatin powder
 2. Explain: Affects the rate of hydrolysis by bromelain
- OR
3. State: pineapple used
 4. Explain: different pineapple may have different concentration of enzymes
 5. State: volume of pineapple juice used
 6. Explain: 1cm³ of 100% pineapple juice and 2cm³ of 50% similar enzyme units

[2]

(iv) State one possible source of error in your investigation and suggest how this can be improved.

1. State: Temperature of set up
2. Suggest: thermostatically controlled water bath to control for temperature during the reaction
3. State: Volume of gelatin gel used in different sample tube is different
4. Suggest: Measure mass / height of gel used

[2]

- (v) Suggest a suitable control experiment to show that bromelain is the cause of liquification of gelatin.

Boiled and cooled bromelain solution / pineapple juice (all other variable remain constant) [1]

R: distilled water

- (vi) Rennin is a protease. However, it does not cause gelatin to liquefy.

Suggest and explain why rennin does not breakdown gelatin.

1. Rennin active site does not have a complementary 3D conformation/charge to the (amino acids of) gelatin/collagen [2]
2. Cannot bind to form Enzyme-Substrate complex

- (b) A student performed a similar experiment at 30 °C to investigate the effect of different pH on bromelain activity. The pineapples were first blended into a juice, and pH buffer added. The time taken for gelatin with the different juice and pH buffer to set were recorded. If the gelatin did not set after 600 seconds, the time recorded will be 'more than 600'.

The results are shown in Table 1.1.

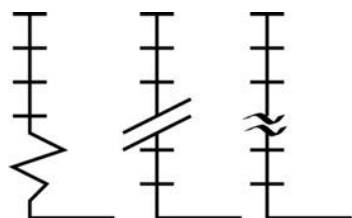
Table 1.1

pH of solution	time taken for gelatin to set / s
1	420
3	505
5	510
7	more than 600
13	470

- (i) Plot a graph of your results from Table 1.1 in the grid below.

G – dot to dot line graph + no extrapolation of line drawn beyond highest plotted point [Reject: best-fit, bar graph] [4]

A – correct axis labels + regular axis marking (y-axis: time taken for gelatin to set / s , x-axis: pH of solution) [Reject axis breaks]



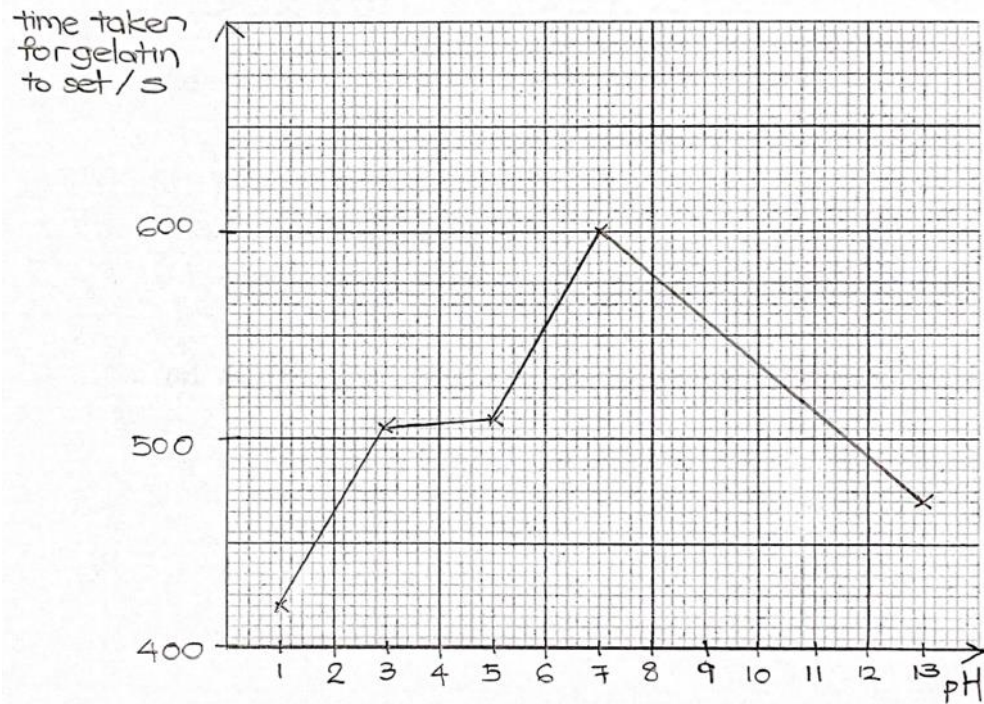
Note: for bar graph, A can be awarded if all criteria for bar graph is fulfilled: Y-axis must start from zero, X-axis has no arrow head

P – correct points drawn [Reject random plotting of >600 for pH 7]

Note: for bar graph, P can be awarded if bars are of equal width and interval, correct height.

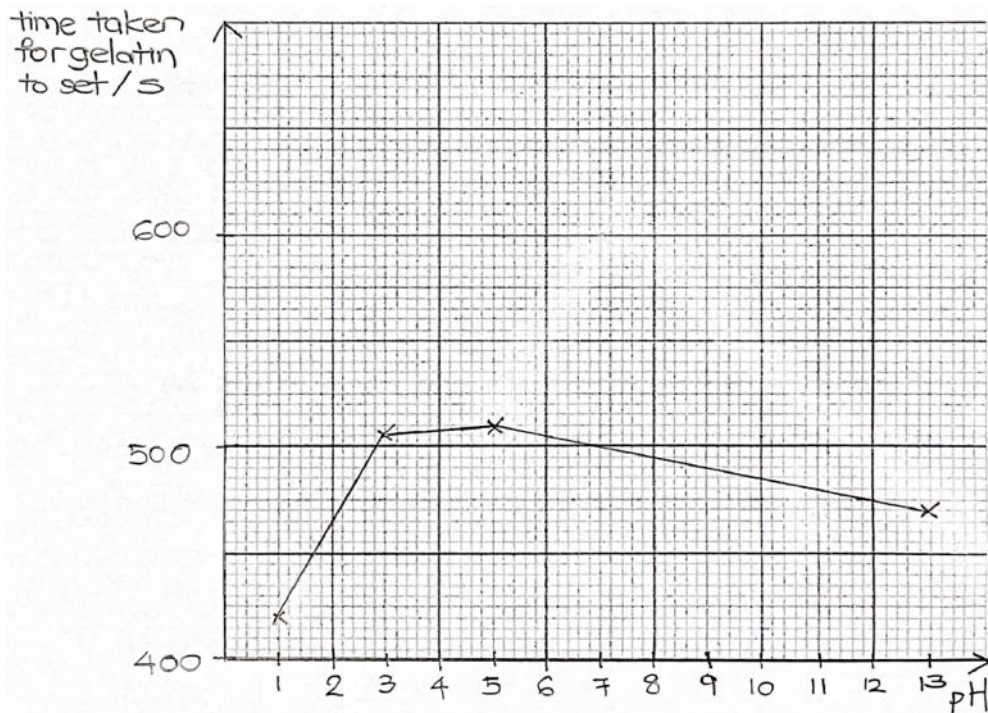
S – Scale $\frac{3}{4}$ of grid used (10 small square rep 40 if students draw 5 points)

Note: for bar graph, scale from leftmost edge of first bar to rightmost edge of last bar is at least $\frac{3}{4}$

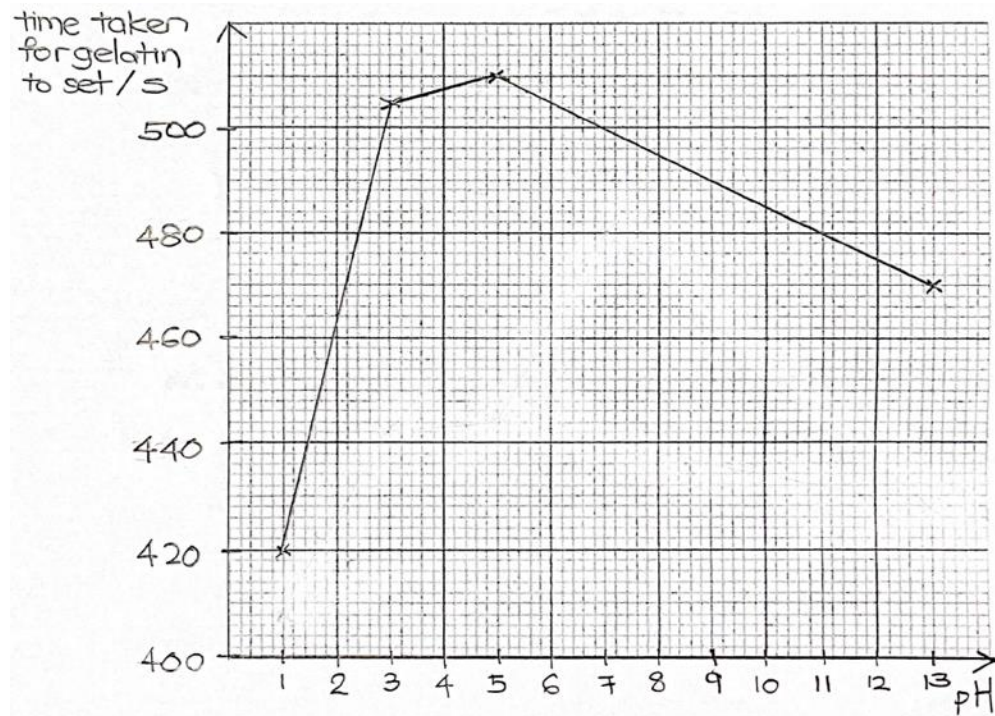
Example 1:

G – 1; A – 0; P – 1 (accept plot >600 as 600) ; S – 0;

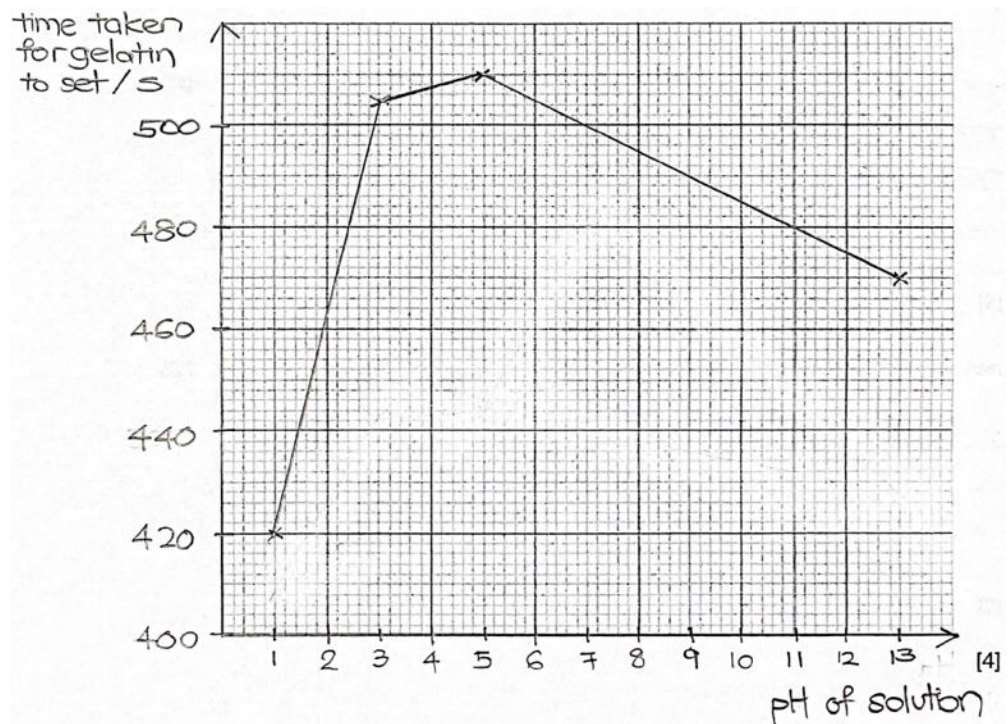
[4]

Example 2:

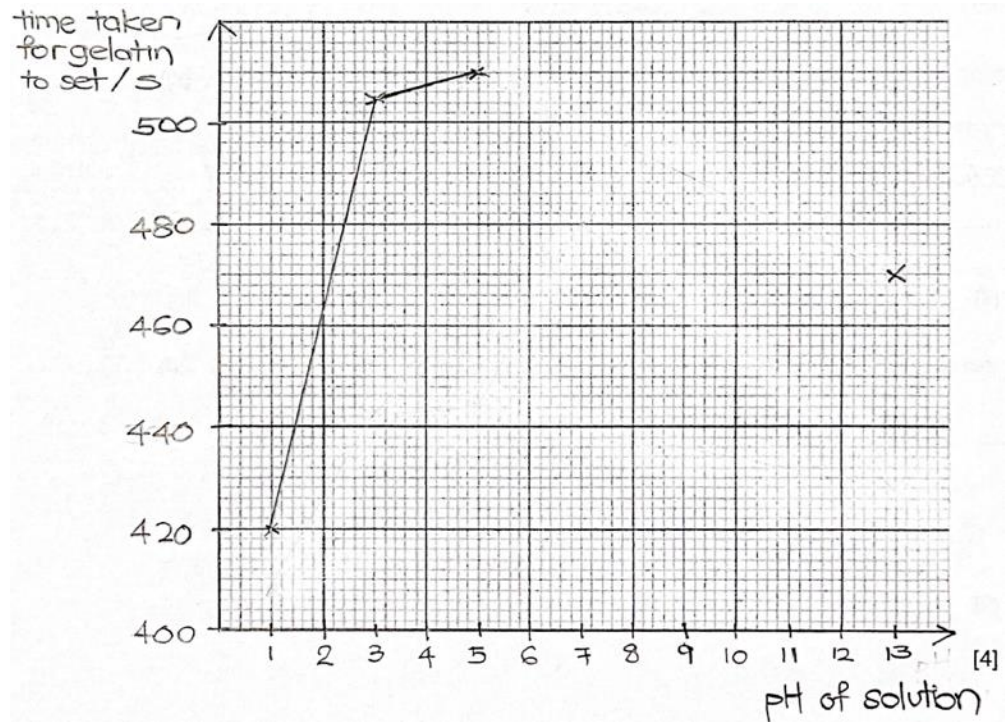
G – 1; A – 0; P – 1 (accept no plot for >600); S – 0;

Example 3:

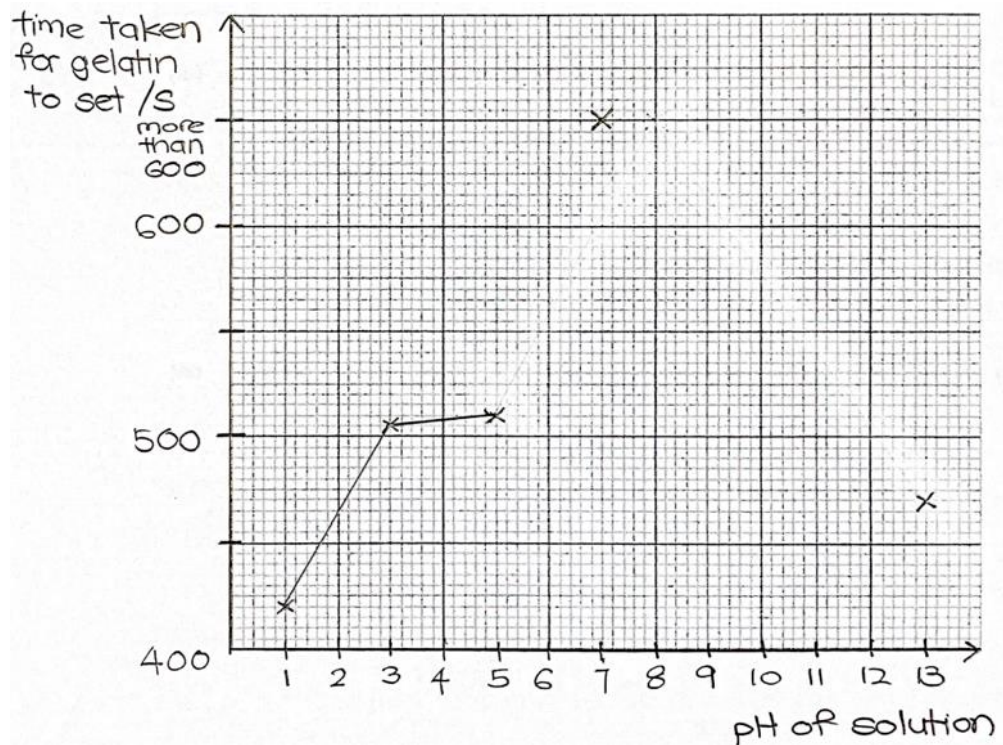
G – 1; A – 0 (x-axis label not as per Table) ; P – 1; S – 1;

Example 4:

G – 1; A – 1; P – 1; S – 1

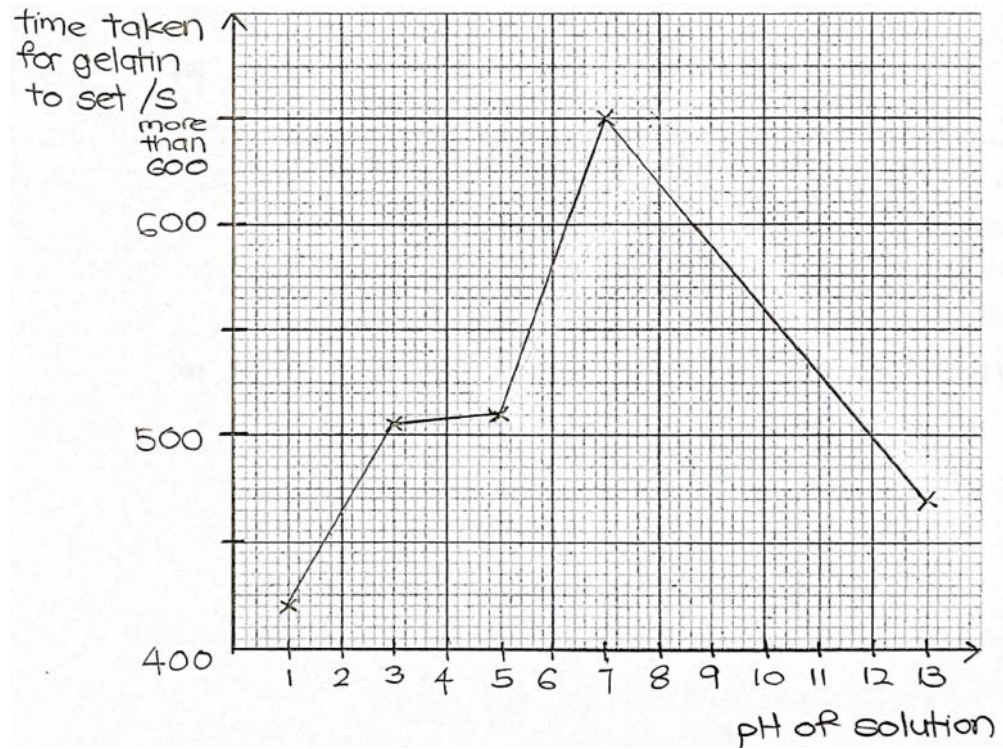
Example 5:

G – 1 (accept dot-2-dot plot from pH1 to pH 5); A – 1; P – 1; S – 1;

Example 6:

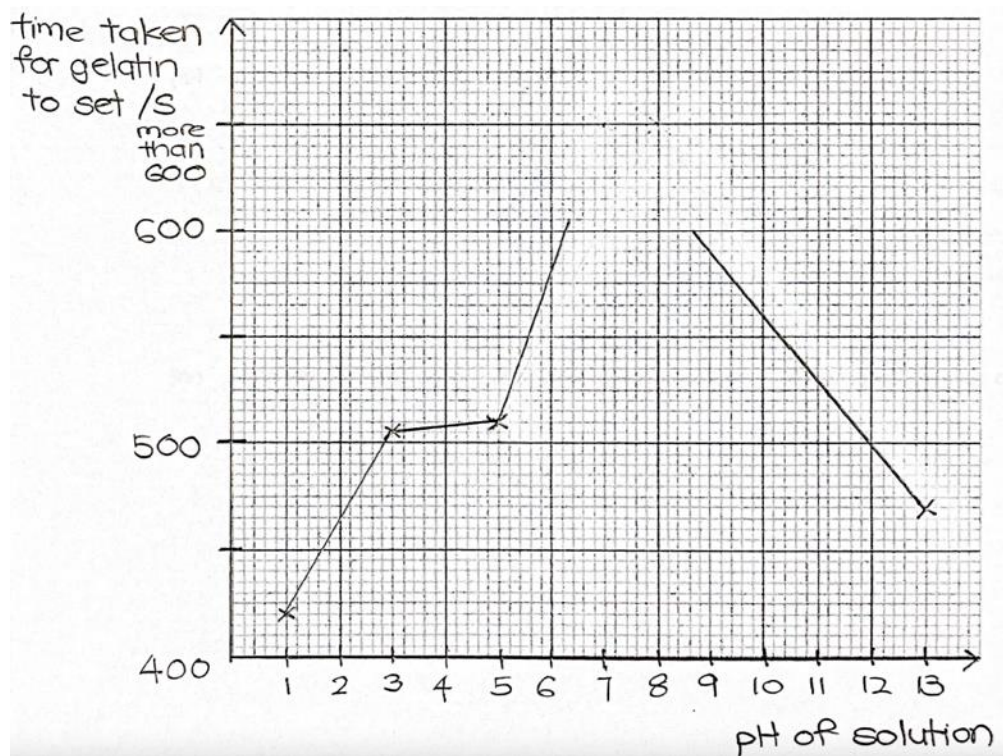
G – 1; A – 0 (irregular axis marking); P – (random point); S – 1;

Example 7:



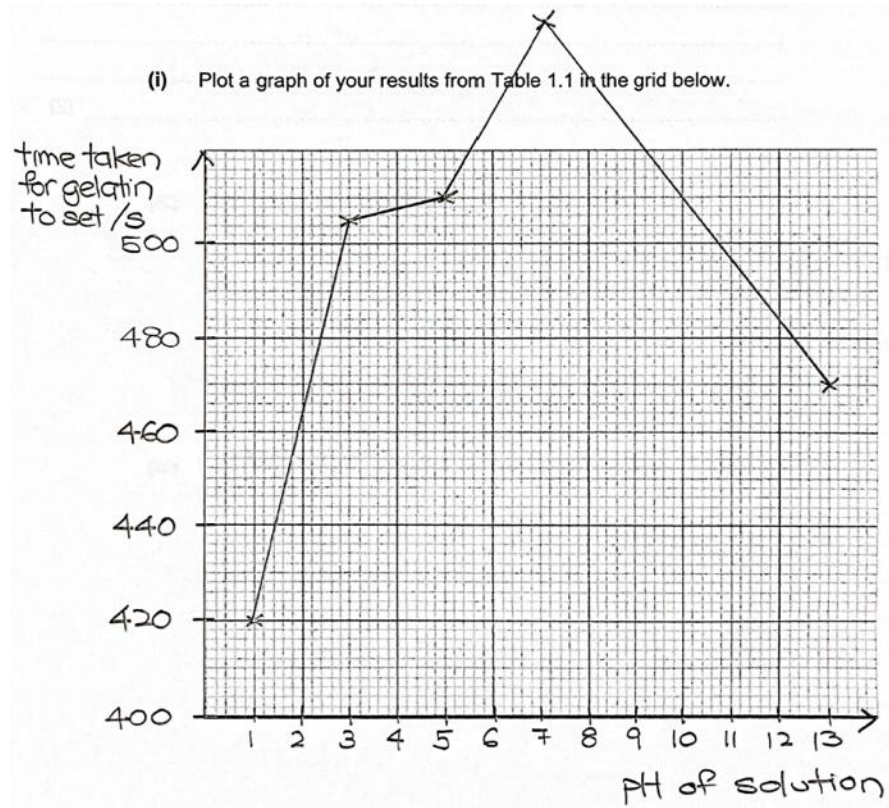
G – 1; A – 0 (irregular axis marking); P – 0 (random plotting); S – 1;

Example 8:



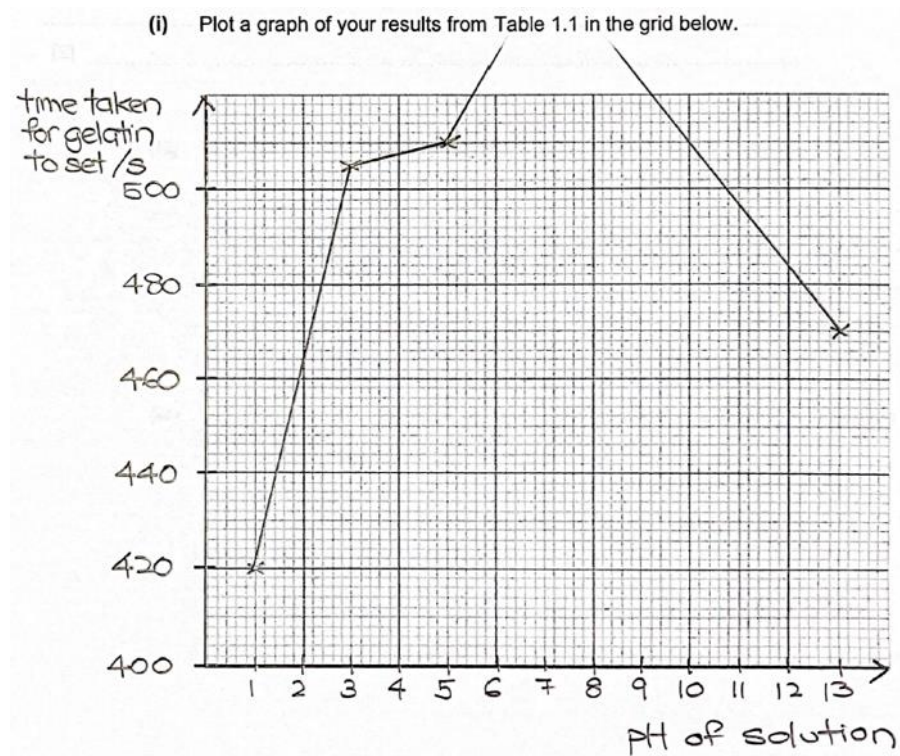
G – 0 (extrapolation); A – 0 (irregular marking on axis); P – 1; S – 0

Example 8:



G – 1; A – 1; P – 0 (random point); S – 1;

Example 9:



G – 0; A – 1; P – 1; S – 1;

(ii) With reference to Table 1.1, explain the effect of pH on the rate of bromelain activity.

1. As pH increases from 1 to 7, time taken for gelatin to set increases from 420 s to more than 600 s and [4]
2. time taken for gelatin to set decreased to 470 s at pH 13
3. At **pH 7**, bromelain amino acids acquire the **correct charges**, hence **conformation & charge of active site becomes most complementary to amino acids of collagen**
4. Ref: Below and above pH 7, change in pH altered the charges of the acidic and basic R-groups of amino acids of the enzyme
5. **disruption of ionic bonds and hydrogen bonds** that are involved in maintaining the specific three-dimensional structure of the enzyme □ **bromelain denatured / non-functional**

(iii) Suggest why the enzyme bromelain does not digest the proteins in the stomach when fresh pineapple is eaten.

Ref: pH of stomach is less than or equal 3 / very low pH, bromelain denatured [1]
/ change in 3D conformation

[Total: 25]

- 2 During this question you will require access to a microscope to study plant adaptations. Adaptation may be defined as the fitness or adjustment of an organism to its environment or habitat by changing morphological, anatomical, or physiological features.

- (a) The moses-in-the-cradle plant is low plant that is used to being shaded by tall trees. The leaves are dark green on top and purple on the underside.

You are provided with a section of the leaf, **M**, in a petri dish labelled **M**.

You are not expected to be familiar with this specimen.

You are required to:

- make a microscope slide of transverse sections of this leaf
- record observations of the tissues.

Proceed as follows.

- 1 Identify the part of **M** that is at least 18 mm wide. Using a penknife, make transverse sections of **M** that are **less than 0.5 mm** in thickness. It is important to make very thin sections of the leaf for details of the tissues to be clearly observable under the microscope.



- 2 Place two of the thinnest and complete transverse sections on a clean slide. Label this microscope slide **M1**.
- 3 Observe the transverse sections using the low-power objective lens of your microscope.
- 4 Focus on a region of the transverse section where vascular tissues are observable.

- (i) Draw a plan diagram of one transverse section.

A plan diagram shows the arrangement of different tissues. Your drawing should show the correct shape and proportion of the different tissues.

No cells should be drawn.

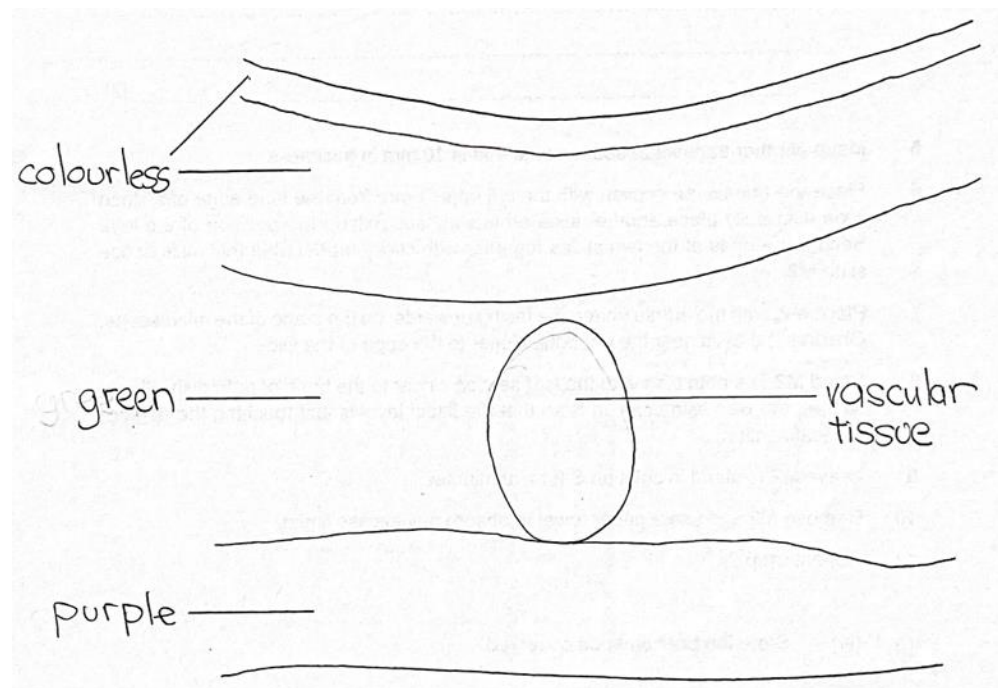
C – clean, continuous lines;

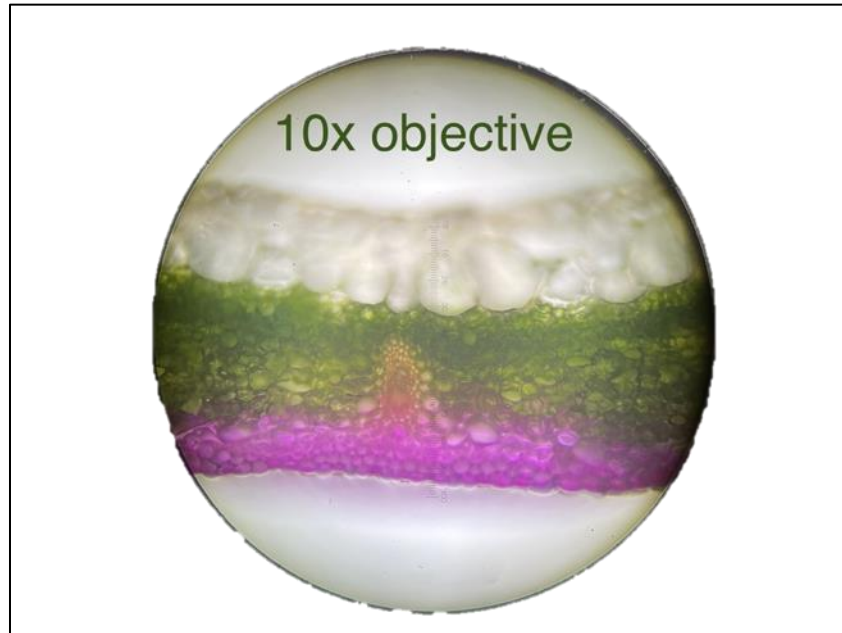
[4]

S – width > 11cm + height > 10cm;

P – proportion of colourless (top two layers - epidermis is $\frac{1}{5}$ of the colourless layer) : green (middle layer) : purple (bottom layer) is approximately 3 : 2 : 1 or 2 : 2 : 1 or 2 : 3 : 1

D – shows four layers + vascular tissue in the green (middle layer), “open-ended” ;



Sample for Drawing

(ii) On your drawing in (a)(i):

- add a labelling line and the label 'vascular tissue' to identify the region that is vascular tissue
- add appropriate labelling lines and labels to identify the colours of all other tissues.

1. labels 'vascular tissue' appropriately; [reject if vascular tissue is located in colourless layer]
2. labels middle layer as 'green/ dark green' + bottom layer as 'purple'

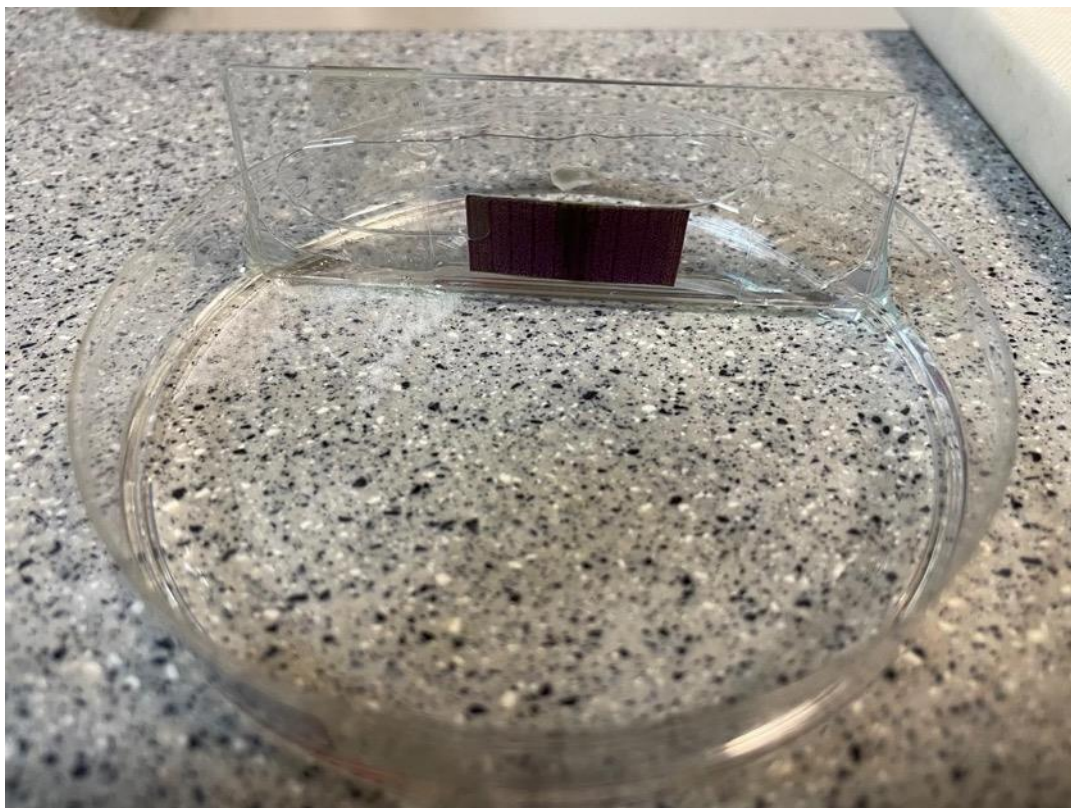
[3]

3. labels top layer as 'colourless/ no colour' [reject: transparent/
clear/white/light green]

- (iii) Suggest how the purple pigmentation may have helped the plant adapt to its environment.

1. Reflects blue/red light from underside of leaf into mesophyll layer;
 2. Allows chlorophyll to absorb more light that may not have otherwise been absorbed;
- [2]

- 5 Make another transverse section of **M** that is 10 mm in thickness.
- 6 Place the transverse section with the cut edge 1 mm from the long edge of a clean slide. Carefully place another slide on top without shifting the position of the leaf. Secure the ends of the two slides together with sticky tape. Label this microscope slide **M2**.
- 7 Place **M2**, with the purple underside facing upwards, on the stage of the microscope. Observe the cells near the cut edge closer to the edge of the slide.
- 8 Stand **M2** in a petri dish with the leaf section closer to the base of petri dish. Slowly fill the petri dish with solution **S** so that the liquid level is just touching the edge of the leaf section.



- 9 Leave **M2** to stand in solution **S** for five minutes.
- 10 Remove **M2** and use a paper towel to absorb any excess liquid.
- 11 Repeat **step 7**.

- (iv) State the phenomenon observed.

plasmolysis

[1]

- (v) Select **three** adjacent epidermal cells that showed changes in **step 11**.

Each cell touches only one other cell.

Make a large drawing of this line of **three** cells.

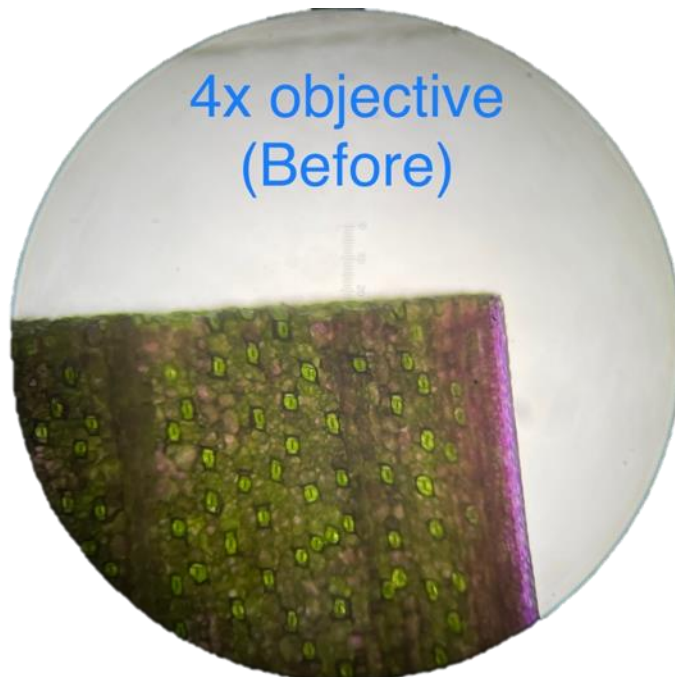
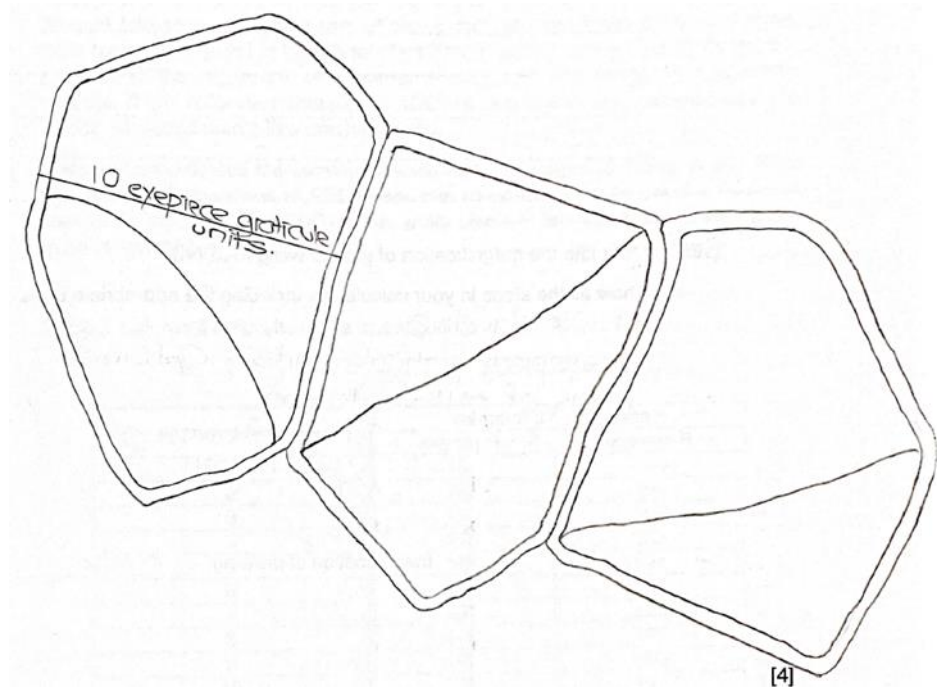
C – clean, continuous lines; [reject if sides of cell drawn with ruler]

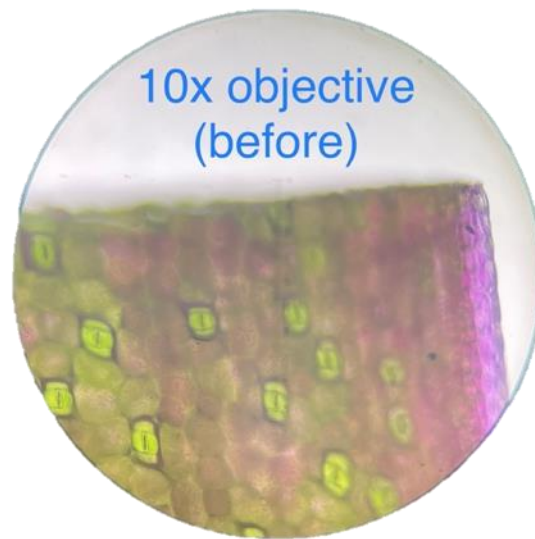
[4]

S – width > 11cm + height > 10cm;

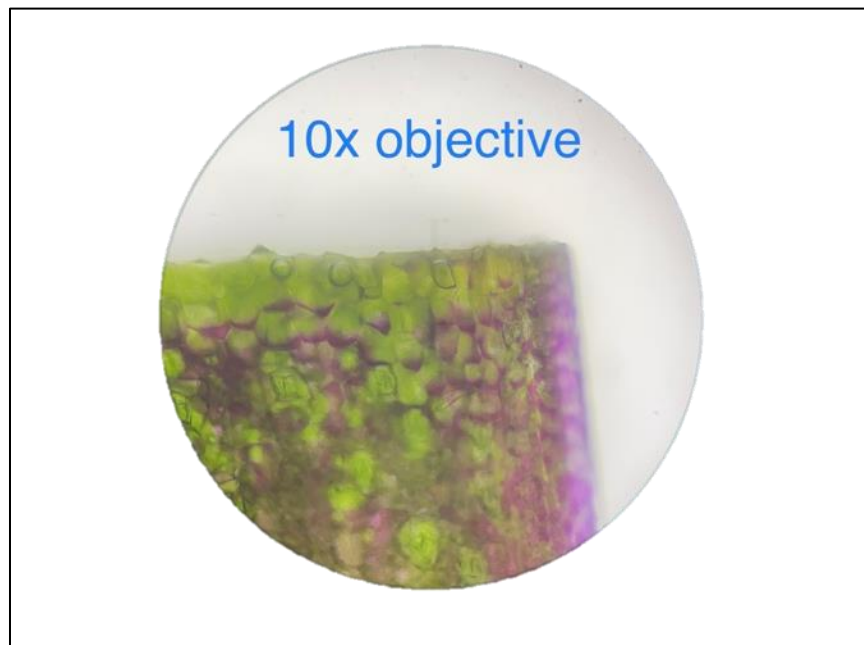
P – cell wall is between four to six $\frac{1}{2}$ of cell + even cell wall thickness;

D – shows shrinkage of cytoplasm (R: presence of nucleus)





Sample for Drawing



(vi) On your drawing in (a)(v):

- draw a line to indicate the width of **one** cell
- write on the line the width of this cell in eyepiece graticule units.

1. Follows instructions and cell width is _____ eyepiece graticule units;

[1]

- (vii) State the objective lens used for making the drawing in (a)(v), and explain how you will determine the width of the cell in μm .

objective lens **10x**; [Reject: 40X, will crack slide] ;

explanation Allow for ECF

Any two:

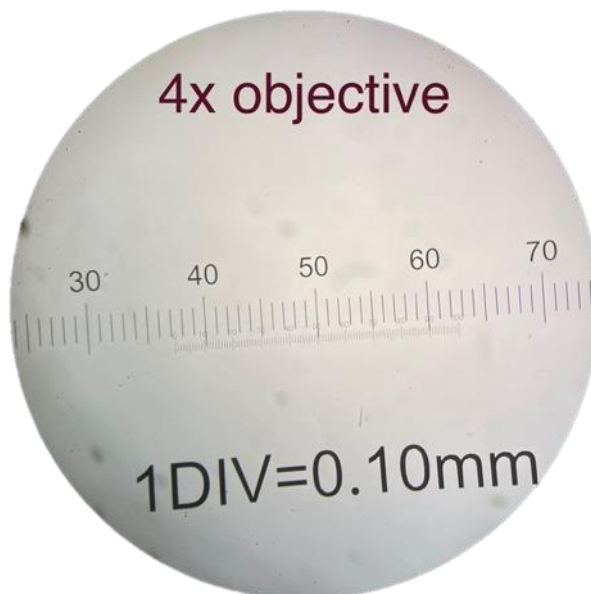
[3]

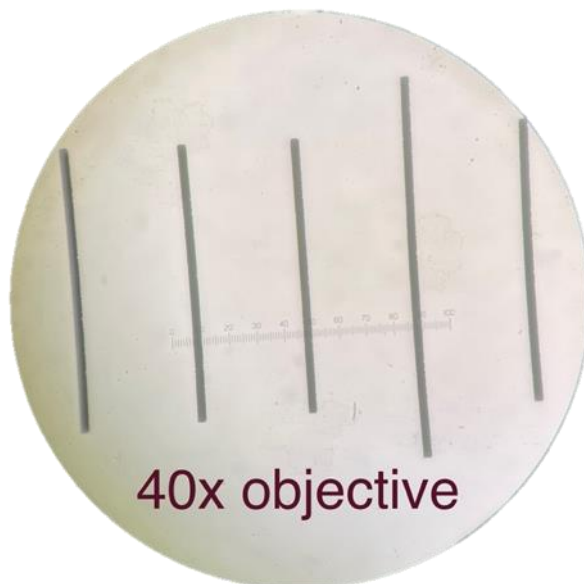
1. Align the stage micrometer scale with eyepiece graticule divisions;
2. Count number of eyepiece graticule divisions in 0.1mm and calculate the width of each division;

compulsory

3. Multiply the number of eyepiece graticule divisions by width of each division;

[reject: count the number of eyepiece graticule units occupied by width of cell because mark point awarded in (vi)]





- (viii) Calculate the magnification of your drawing in (a)(vi).
Show all the steps in your calculation, including the appropriate units.

No ECF for 40X

ECF for if length is drawn on (vi)

Check actual length of drawing for marking point W. No mark awarded for W if actual length measured is wrong.

W – working presented with short annotation for mathematical manipulation of numbers;

A – final answer presented to _____X/ X_____ (2 or 3 sf) [Note: final answer should be in the hundreds] ;

Example:

At 10x objective,

10 eyepiece graticule units = 0.1 mm

Width of cell = 0.1 mm

$$\text{Magnification} = \frac{\text{width of cell on drawing}}{\text{width of actual cell}} = \frac{59}{0.1} = 590X$$

magnification of drawing:

.....
[2]

- (b) Global warming and associated precipitation changes will negatively impact on many agricultural ecosystems. Major food production areas are expected to experience reduced water availability and increased frequency of drought over the coming decades.

Almost all water used for plant growth is lost to the atmosphere by transpiration through stomatal pores on the leaf epidermis. Several approaches to improve drought tolerance and water-use efficiency through the modification of stomatal traits have been tested in the model plant *Arabidopsis thaliana*. The *SDD1* gene is involved in the regulation of stomatal density and distribution in *Arabidopsis thaliana*. A loss of function mutation in *SDD1* gene was shown to increase stomatal density in mutant plants in a previous study.

A student investigated the stomatal density in two lineages of plants, **A** and **B**, to find out if overexpression of *SDD1* gene can decrease stomatal density. Plants in lineage A overexpressed *SDD1* gene, while plants in lineage B expressed *SDD1* gene at normal levels.

Table 2.1 shows the results of the investigation.

Table 2.1

sample number	stomatal density / mm ⁻²	
	lineage A	lineage B
1	181	190
2	182	191
3	190	192
4	178	193
5	184	194
6	183	194
7	181	196
8	179	197
9	190	190
10	179	192
mean ($\bar{x}$)	182.7	193.1
standard deviation (s)	4.27	2.81
variance (s^2)	18.23	7.90

- (i) Complete Table 2.1 by calculating the variance (s^2) for the stomatal density of two lineages: lineage **A** and lineage **B**.

[1]

- (ii) A t -test can be used to determine whether there is any significant difference between the stomatal density in lineage **A** and lineage **B**.

Calculate the value of t and the number of degrees of freedom, using these formulae:

$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}} \quad v = n_1 + n_2 - 2$$

key to symbols

s = standard deviation

$\bar{x}$ = mean

n = sample size (number of observations)

v = degrees of freedom

Show your working.

$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}} = 3.34$$

W – shows figures correctly substituted into formula;

A – gives correct t and degrees of freedom

value of t = **6.43**

Accept 6.44 only if (i) reported
18.2 for lineage A

number of degrees of freedom = **18**

[2]

(iii) State the null hypothesis.

There is no significant difference between the means of lineage A and lineage B; [1]

Table 2.2 shows the critical values for t at different probabilities and degrees of freedom for one-tailed test and two-tailed test.

Table 2.2

degrees of freedom	probability, p , for one-tailed test			
	0.25	0.05	0.025	0.005
	probability, p , for two-tailed test			
	0.5	0.1	0.05	0.01
1	1.00	6.31	12.71	63.66
2	0.82	2.92	4.30	9.92
3	0.76	2.35	3.18	5.84
4	0.74	2.13	2.78	4.60
5	0.73	2.02	2.57	4.03
6	0.72	1.94	2.45	3.71
7	0.71	1.89	2.36	3.50
8	0.71	1.86	2.31	3.36
9	0.70	1.83	2.26	3.25
10	0.70	1.81	2.23	3.17
11	0.70	1.80	2.20	3.11
12	0.70	1.78	2.18	3.05
13	0.69	1.77	2.16	3.01
14	0.69	1.76	2.14	2.98
15	0.69	1.75	2.13	2.95
16	0.69	1.75	2.12	2.92
17	0.69	1.74	2.11	2.90
18	0.69	1.73	2.10	2.88
19	0.69	1.73	2.09	2.86
20	0.69	1.72	2.09	2.85

- (iv) Use Table 2.2 and your answers to (b)(ii) to decide whether the null hypothesis should be accepted or rejected.

Explain your answer.

accept or reject null hypothesis **Reject;**

explanation **calculated t is greater than critical value of 1.73; (must quote critical t value)**

[2]

- (v) On the grid provided, plot a graph of the means shown in Table 2.1.

G – bar graph ;

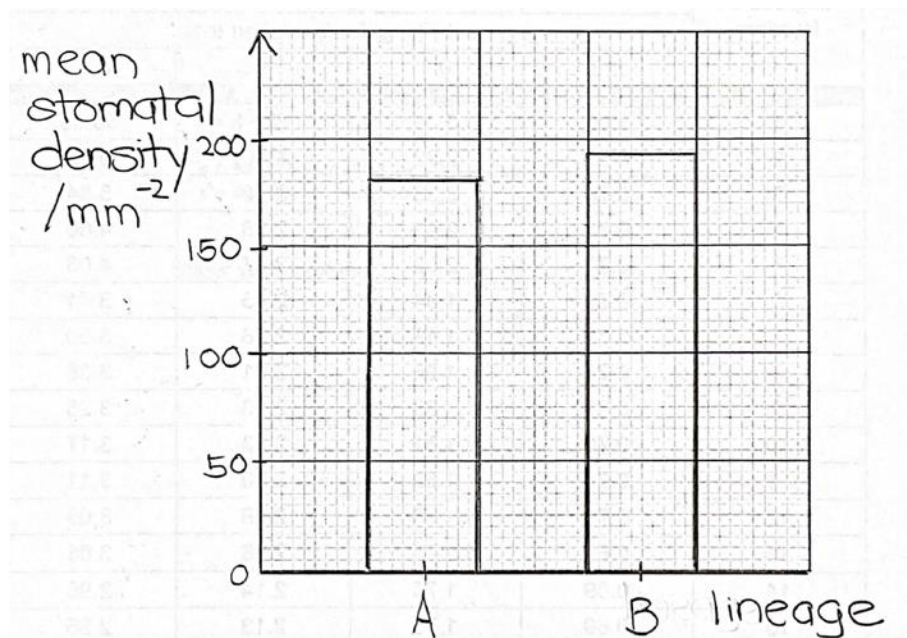
[3]

A – shows average stomatal density (y-axis + start from zero), lineage (x-axis + no arrow head) ;

P – bars are of equal width and equal interval and plotted accurately;

S – not assessed

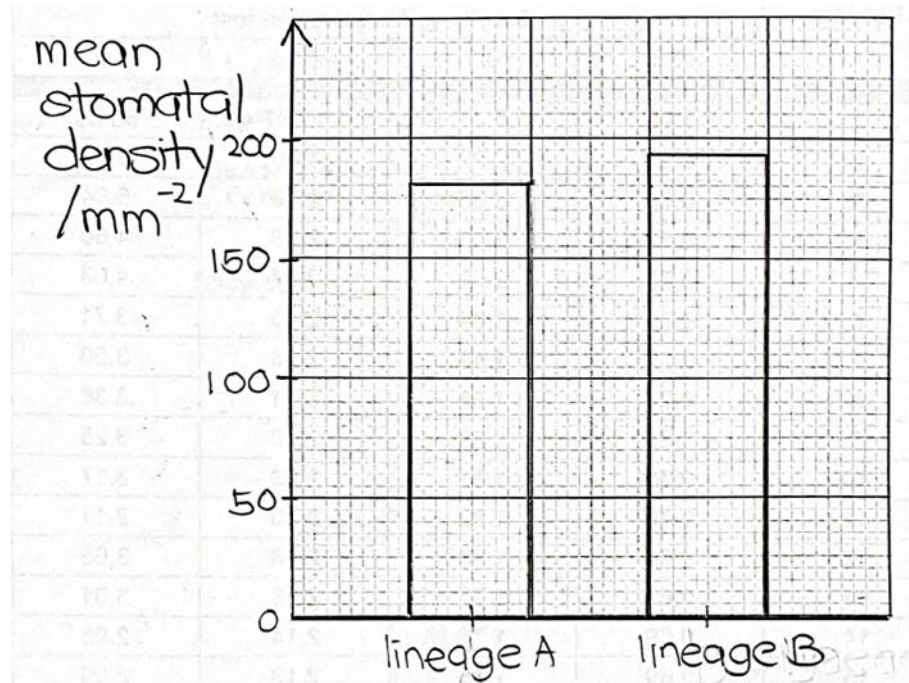
Example 1



[3]

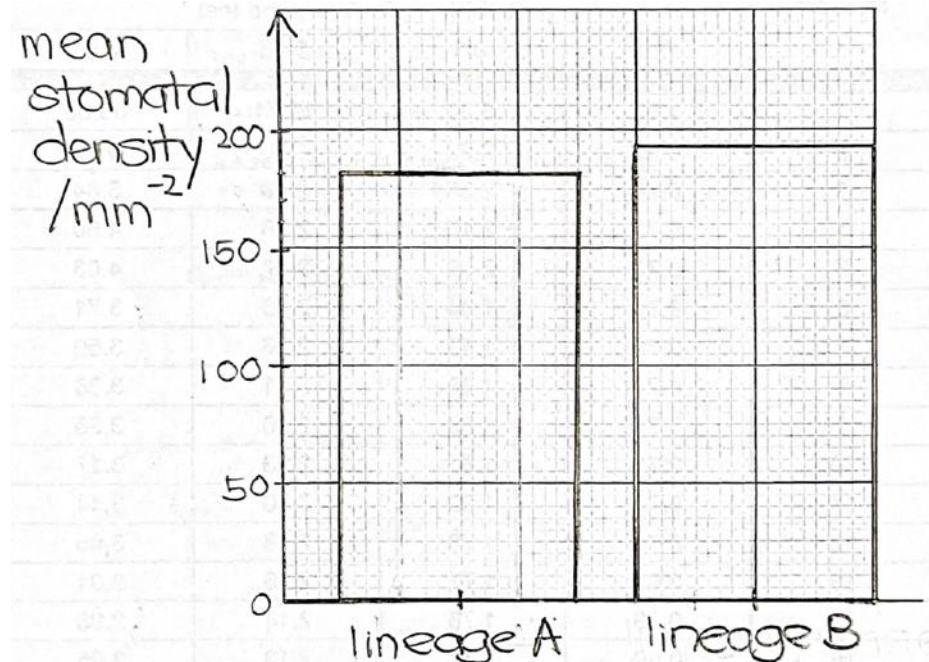
G – 1; A – 1; P – 1; S – 0 (leftmost edge of first bar to rightmost edge of last bar covers less than $\frac{3}{4}$)

Example 2



G – 1; A – 1; P – 1; S – 0 (leftmost edge of first bar to rightmost edge of last bar covers less than $\frac{3}{4}$)

Example 3



G – 1; A – 1; P – 1; S – 1 (leftmost edge of first bar to rightmost edge of last bar covers more than $\frac{3}{4}$)

(vi) Explain if the graph in (b)(v) is sufficient to determine the stomatal density in lineage **A** is significantly different from lineage **B**.

1. Not sufficient + bars of the similar height; [1]
2. Not sufficient + no indication of variance/ SD;
3. Not sufficient + difference in height is dependent on scale used for axis;

Reject:

(R!) small sample size (not something evident from the graph)

[Total: 30]